

Investment Appraisal Consolidation Activity

Pearson Edexcel IAL Business, Unit 3: Investment appraisal (payback, ARR and NPV). Case study: Greggs plc, Kettering vs Derby.

Name: _____	Class: _____	Date: _____
-------------	--------------	-------------

Instructions: Read Extracts A to E. Complete Section A one method at a time, then Section B. Use the 10% discount factors in Extract D. Show all workings. Round payback to the nearest month, ARR to one decimal place and NPV to two decimal places (£m).

The case study: Greggs plc

Extract A: Greggs plc

Greggs is a UK food-to-go retailer with over 2,600 shops, and its strategic growth plan aims to take the estate from around 2,500 shops to well over 3,000. Its 2025 results described 2025 as the peak year of a multi-year investment programme, with capital expenditure of around £300 million, mostly on new supply chain sites. Underlying return on capital employed (ROCE) fell from 20.3% in 2024 to 16.0% in 2025 because of this planned increase in capital employed. Greggs made structural cost savings of £13.0 million in 2025, and says free cash generation should rise as its investment needs fall.

Extract B: Project A, Kettering National Distribution Centre

Greggs bought the land for its new National Distribution Centre (NDC) at Symmetry Park, Kettering for £30 million in January 2025 and expects a further £105 million of cash-funded capital investment, a total of £135 million. The site will store, pick and distribute chilled and ambient goods, with an automated cold store and automated picking down to shop level. It is expected to be fully operational in 2027 and will allow around 900 further net new shops, which is six to nine years of expansion at recent opening rates of 100 to 150 net new shops a year.

Extract C: Project B, Derby frozen production and logistics site

Greggs has leased a 23-acre plot at SmartParc SEGRO in Spondon, Derby, for a frozen production and logistics facility due to be fully operational by the end of 2026. It will add sweet and savoury production lines, frozen storage and fully automated robotic picking for shops, and is expected to create up to 600 jobs. Extra capacity is available for future production lines and new products. Construction of the building was carried out by the landlord, so Greggs' own spending is on fit-out, equipment and automation.

Extract D: Forecast net cash flows and discount factors (illustrative)

Greggs has not published forecast cash flows for these sites. The figures below are invented for this exercise. Assume Greggs can fund only one project first, both projects last five years and there is no residual value.

Year	Project A: Kettering NDC net cash flow (£m)	Project B: Derby site net cash flow (£m)	Discount factor at 10%
0	(135)	(120)	1.000
1	50	20	0.909
2	45	30	0.826
3	35	40	0.751
4	25	50	0.683
5	20	60	0.621

Extract E: Other information

- Project A serves the existing shop network sooner: it relieves pressure on Greggs' current regional distribution centres and lets new shops open without extra capacity constraints. Cash flows are front-loaded but fall each year as the network matures.
- Project B adds production capacity and frozen logistics for future growth, including new product lines. Cash flows start low while the site ramps up, then rise as production lines fill.
- Both sites use advanced automation, so delays in commissioning or technology problems could hold back cash flows.
- Greggs' profitability is under pressure from rising labour and food costs, and ROCE has fallen.
- Greggs is committed to sustainability targets, and both sites are designed with energy efficiency and lower emissions in mind.

Sources: Greggs preliminary results and announcements (Investegate RNS; Greggs 2025 results, 3 March 2026; Greggs Derby announcement). Adapted for teaching; all forecast cash flows and Derby costs are illustrative.

Section A, Task 1: Payback

Project B is already completed for you. Calculate Project A, then compare.

Year	Project A cumulative net cash flow (£m)	Project B cumulative net cash flow (£m)
0	(135)	(120)
1		(100)
2		(70)
3		(30)
4		20
5		80

*Project B payback: after Year 3, £30m is still to be recovered. $\text{£}30\text{m} \div \text{£}50\text{m} \times 12 \text{ months} = 7 \text{ months}$, so payback is **3 years 7 months**.*

1a. Calculate the payback period for Project A. Give your answer in years and months. **(3 marks)**

.....

.....

.....

1b. State which project payback favours and by how many months. **(2 marks)**

.....

.....

1c. Explain one benefit and one drawback of using payback to appraise these projects. Refer to Greggs in your answer. **(4 marks)**

.....

.....

.....

.....

.....

.....

Section A, Task 2: Average rate of return (ARR)

Project A is already completed for you. Calculate Project B, then compare.

Calculation	Project A: Kettering (£m)	Project B: Derby (£m)
Total net cash inflows (Years 1 to 5)	175	
Less initial investment	(135)	
Total net return	40	
Divide by project life (5 years) = average annual profit	8	
ARR = average annual profit ÷ initial investment × 100	5.9%	

Project A: $(£175m - £135m) \div 5 = £8m$ a year. $£8m \div £135m \times 100 = 5.9\%$ (to one decimal place).

2a. Complete the Project B column. Give ARR to one decimal place. **(4 marks)**

.....

.....

.....

2b. State which project ARR favours. Explain what the two ARR figures suggest about how profitable each project is relative to its cost. **(2 marks)**

.....

.....

2c. Explain one benefit and one drawback of using ARR to appraise these projects. Refer to Greggs in your answer. **(4 marks)**

.....

.....

.....

.....

.....

.....

Section A, Task 3: Net present value (NPV)

Project B is already completed for you. Calculate Project A, then compare. Use the discount factors in Extract D and give answers in £m to two decimal places.

Year	Net cash flow (£m)	Discount factor	Project A: present value (£m)	Project B: present value (£m)
1	A: 50, B: 20	0.909		18.18
2	A: 45, B: 30	0.826		24.78
3	A: 35, B: 40	0.751		30.04
4	A: 25, B: 50	0.683		34.15
5	A: 20, B: 60	0.621		37.26
Total present value				144.41
Less initial investment			(135)	(120)
NPV				+24.41

3a. Complete the Project A column and calculate its NPV. **(4 marks)**

.....

.....

.....

3b. State which project NPV favours. Explain what a positive NPV means for Greggs. **(2 marks)**

.....

.....

3c. Explain one benefit and one drawback of using NPV to appraise these projects. Refer to Greggs in your answer. **(4 marks)**

.....

.....

.....

.....

.....

.....

Section B: Bringing the methods together

Task 4: Results grid (3 marks). Copy your results from Section A into the table and tick the project each method favours.

Measure	Project A: Kettering NDC	Project B: Derby site	Method favours
Payback			
ARR			
NPV			

Task 5. Using the data in Extracts C and D, assess the usefulness of payback to Greggs when deciding between Project A and Project B. **(8 marks)**

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for Task 5 = 8 marks)

Task 6: Comparing the two projects (12 marks)

Using the data in Extracts B, C, D and E, together with NPV and ARR, assess which project Greggs should choose: Project A, the Kettering National Distribution Centre, or Project B, the Derby frozen production and logistics site. **(12 marks)**

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for Task 6 = 12 marks)

Task 7: Evaluation (20 marks)

Evaluate which investment is likely to be the most successful for Greggs: Project A, the Kettering National Distribution Centre, or Project B, the Derby frozen production and logistics site. Use investment appraisal and other factors in your answer. **(20 marks)**

Plan first (5 minutes). Complete the grid before you write.

Evidence	Analysis (compare A and B directly)	However	Mini judgement
Payback			
ARR			
NPV			
Qualitative factors from Extract E (growth, risk, automation, ROCE, sustainability)			

Level 4 checklist

- Compare figures directly, using the same measure for both projects.
- Develop the consequence for Greggs, not just the number.
- Challenge each argument with “However”.
- Add a mini judgement in every paragraph: which evidence carries more weight, and why.
- Finish with a conclusion that answers: Which project? Why is it stronger? Why reject the other? What does the decision depend on? What extra evidence is needed?

Recommended writing time: 30 to 35 minutes.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for Task 7 = 20 marks)

Stretch it. Would your judgement change if:

- Project B's cash flows were 15% lower than forecast because of commissioning delays? Recalculate its NPV.
- Project A's cash flows were 10% lower because shop openings slowed? Recalculate its NPV.
- The cost of Project B rose by 15% (£120m to £138m)? Recalculate its NPV.
- Greggs' cost of capital were 12% rather than 10%? Which factors would you need to re-check?